HANDS-ON ACTIVITY

# Build a Traffic Light!

A paper craft + partner game that teaches signal logic by doing it yourself

GRADES K-5 | MIDDLE SCHOOL | 25-35 MIN | SCISSORS + BRAD

## What You'll Need

- This printed sheet (page 2 has the craft template)
- Scissors
- A brass paper fastener (brad) — or a sticker dot + tape as a substitute
- Crayons or markers: red, yellow, green
- A partner!

## Part 1 — Build It

- Color the three circles on the template: top = red, middle = yellow, bottom = green.
- Cut out the traffic light housing (the black rounded rectangle) and the small spinning arrow on page 2.
- Poke the brad through the center dot of the arrow, then through the dot in the middle of the housing, and fold the prongs flat in back.
- Spin the arrow to point at red, yellow, or green — that's your working traffic light!

### How Real Signals Work

Real traffic lights don't just flip randomly; they follow a **cycle**: green → yellow (a few seconds warning) → red → back to green. Your spinner should always move in that order, just like the real thing.

## Part 2 — Play 'Be the AI'

Now use your traffic light to play a partner game that mimics how the AI Adaptive mode in the website's simulation makes decisions.

| Step | What Happens |
| --- | --- |
| 1 | One partner is the 'Caller.' The other holds the traffic light. |
| 2 | The Caller rolls a die twice (or picks two numbers 1-6): one for 'cars waiting North-South,' one for 'cars waiting East-West.' |

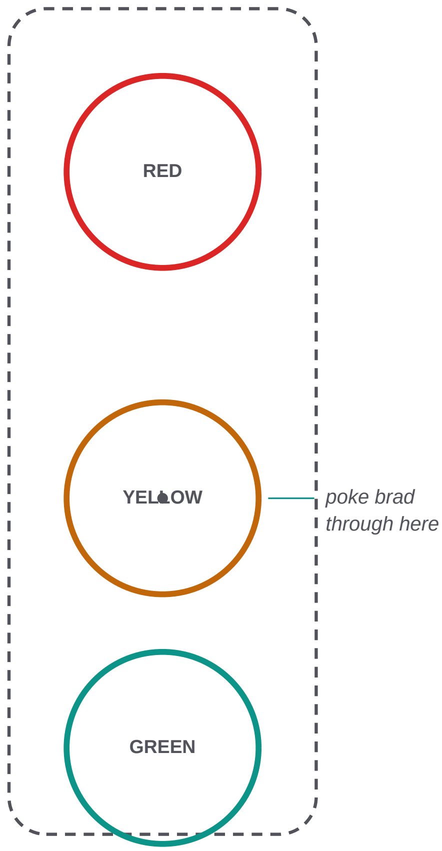
| | |
|---|---|
| 3 | The Light-Holder must decide: which direction has MORE cars waiting? Spin your light to GREEN for that direction's road. |
| 4 | After 3 seconds, spin to YELLOW, then RED, and the Caller rolls again for the next round. |
| 5 | Play 5 rounds, then switch roles! |

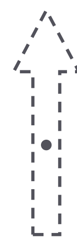
## Think About It

- Was it ever a tough call when both directions had almost the same number of cars?
- A real AI traffic controller has to make this exact decision thousands of times a day. How do you think it decides fairly when the numbers are close?
- What would happen if the Light-Holder ignored the dice and just alternated red/green every turn? (Hint: that's Mode A, the Fixed Timer!)

CRAFT TEMPLATE

# Cut Along the Lines



**Spinner Arrow**

(cut out, poke brad through the dot)